

Question_9:

Code:

```
1  /*
2  Question_9: Find the latest day to take the first train and still finish by D.
3  */
4
5  #include <iostream>
6  using namespace std;
7
8  bool possible(long long first, int n, long long D, long long x[]) {
9      long long day = first;
10
11      for (int i = 1; i < n; i++) {
12          day = ((day + x[i] - 1) / x[i]) * x[i];
13
14          if (day > D)
15              return false;
16      }
17      return true;
18  }
19
20  int main() {
21      int T;
22      cin >> T;
23
24      while (T--) {
25          int n;
26          long long D;
27          cin >> n >> D;
28
29          long long x[1000];
30
31          for (int i = 0; i < n; i++)
32              cin >> x[i];
33
34          long long lo = x[0], hi = D, ans = x[0];
35
36          while (lo <= hi) {
37              long long mid = (lo + hi) / 2;
38              long long first = (mid / x[0]) * x[0];
39
40              if (possible(first, n, D, x))
41                  ans = mid, hi = mid - 1;
42              else
43                  lo = mid + 1;
44          }
45
46          cout << ans << endl;
47      }
48  }
```

```
39         first = x[0];
40
41         if (possible(first, n, D, x)) {
42             ans = first;
43             lo = first + x[0];
44         } else {
45             hi = mid - 1;
46         }
47     }
48     cout << ans << '\n';
49 }
50 return 0;
51 }
```

Output:

Test Case_1:

```
3
3 100
31 71 21
62
4 100
11 16 51 50
44
1 1
1
1
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
3
5 1000
311 171 211 45 32
622
6 1000
111 161 511 510 56 93
111
2 10
9 5
9
PS C:\Users\Prabin Yadav\
```